

Dr. Yijiong Lin

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📍 <https://yijionglin.github.io/>



Education

University of Bristol	Supervisors: Nathan F. Lepora, Efi Psomopoulou,	2020 - 2024
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Ph.D.: Engineering Mathematics

Dandan Zhang

Research Interest: Robot Learning, Dexterous Manipulation, Sim-to-Real, Tactile Sensing, Multi-modality

Guangdong University of Technology

Supervisors: Yisheng Guang, Juan Rojas

2017 - 2020

MSc: Mechanical Engineering

Research Interest: Deep Reinforcement Learning for Robotic Manipulation

Guangdong University of Technology

2013 - 2017

BSc: Mechanical Engineering and Automation (ranking: 1 out of 52)

Academic Positions

University of Bristol	Supervisors: Nathan F. Lepora, Efi Psomopoulou	2024 - Present
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Postdoctoral Research Associate

Research Projects:

1. Democratising Co-Design of Hardware and Control for Robot Dexterity (ARIA)
2. Advancing the Physical Intelligence and Performance of roBOTS towards Human-like Bi-manual Objects MANipulation (Horizon Europe)

Peer-reviewed Publications (selected) (Google Scholar Citation: 363)

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1. K. M. Freud*, **Y. Lin***, et al., “SimShear: Sim-to-Real Shear-based Tactile Servoing”, Conference on Robot Learning (CoRL), 2025. (***Co-first-authorship, Accepted**)
 2. H. Field, M. Yang, **Y. Lin**, et al., “Text2Touch: Tactile In-Hand Manipulation with LLM-Designed Reward Functions”, Conference on Robot Learning (CoRL), 2025. (**Accepted**)
 3. **Y. Lin**, et al., “NeuralTouch: Neural Descriptors for Precise Sim-to-Real Tactile Robot Control,” IEEE/ASME Transactions on Mechatronics (T-Mech), 2025. (*Under-review*)
 4. **Y. Lin**, et al., “Tactile Sensing for Contact-rich Robotic Tasks with Deep Learning Techniques: A Review”, IEEE Sensors Journal, 2025. (*In-progress*)
 5. Z. Wu, **Y. Lin**, et al., “ViTacGen: Robotic Pushing with Vision-to-Touch Generation”, IEEE Robotics and Automation Letters (RA-L), 2025. (*Under-review*)
 6. M. Comi, A. Tonioni, M. Yang, J. Tremblay, V. Blukis, **Y. Lin**, et al., “Snap-it, Tap-it, Splat-it: Tactile-Informed 3D Gaussian Splatting for Reconstructing Challenging Surfaces”, International Conference on 3D Vision, 2025.
 7. M. Comi, **Y. Lin**, et al., “TouchSDF: A DeepSDF Approach for 3D Shape Reconstruction Using Vision-Based Tactile Sensing”, IEEE Robotics and Automation Letters (RA-L), 2024.

8. B. Deng, **Y. Lin**, et al., "Coarse-to-fine Robotic Pushing using Touch, Vision and Proprioception", IEEE Robotics and Automation Letters (RA-L), 2024.
9. M. Yang, C. Lu, A. Church, **Y. Lin**, et al., "Anyrotate: Gravity-invariant in-hand object rotation with sim-to-real touch", Conference on Robot Learning (CoRL), 2024.
10. **Y. Lin**, et al., "Bi-Touch: Bimanual Tactile Manipulation with Sim-to-Real Deep Reinforcement Learning," IEEE Robotics and Automation Letters, 2023. [[paper](#)][[code](#)][[website](#)]
11. **Y. Lin**, et al., "Attention for Robot Touch: Tactile Saliency Prediction for Robust Sim-to-Real Tactile Control", The 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) [[paper](#)][[code](#)][[website](#)]
12. M. Yang, **Y. Lin**, et al., "Sim-to-Real Model-Based and Model-Free Deep Reinforcement Learning for Tactile Pushing", IEEE Robotics and Automation Letters (RA-L), 2023.
13. **Y. Lin**, et al., "Tactile Gym 2.0: Sim-to-Real Deep Reinforcement Learning for Comparing Low-Cost High-Resolution Robot Touch", IEEE Robotics and Automation Letters (RA-L), 2022. [[paper](#)][[code](#)][[website](#)]
14. N. Lepora, **Y. Lin**, et al., "DigiTac: A DIGIT-TacTip Hybrid Tactile Sensor for Comparing Low-Cost High-Resolution Robot Touch", IEEE Robotics and Automation Letters (RA-L), 2022.
15. W. Fan, H. Bo, **Y. Lin**, et al., "Graph neural networks for interpretable tactile sensing", IEEE 27th International Conference on Automation and Computing (ICAC), 2022. (**Best Student Paper Finalist**)
16. J. Lloyd, **Y. Lin**, N. Lepora, "Probabilistic Discriminative Models Address the Tactile Perceptual Aliasing Problem", Robotics: Science and Systems (RSS), 2021.
17. S. Luo, H. Wu, S. Duan, **Y. Lin**, et al., "Endowing robots with longer-term autonomy by recovering from external disturbances in manipulation", Journal of Intelligent & Robotic Systems, 2021.
18. **Y. Lin**, et al., "Invariant Transform Experience Replay: Data Augmentation for Deep Reinforcement Learning", IEEE Robotics and Automation Letters (RA-L), 2020. [[paper](#)][[code](#)][[website](#)]
19. **Y. Lin**, et al. "An Odd-Form Electronic Component Insertion System Based on Dual SCARA", 2018 IEEE International Conference on Robotics and Biomimetics (ROBIO).

Patents and Software Copyright (Selected)

1. **Y. Lin**, et al., "Robot Repair Method and Device," Chinese Invention Patent, Patent No.: ZL201910671929.X, Publication No.: CN110370317B, Date of Authorization: January 11, 2022.
2. Y. Guan (Supervisor), **Y. Lin**, et al., "A Robot End Magnetic Adsorption Module," Chinese Invention Patent, Patent No.: ZL201710044899.0, Publication No.: CN106625739B, Date of Authorization: May 16, 2023.
3. Y. Guan (Supervisor), **Y. Lin**, et al., "Intelligent Control Software for Ultra-High Pressure Water Jet Cutting Platform," Software Copyright Registration No.: 2017SR715521, Date of Authorization: December 24, 2017.

Project Experience

1. ARIA Research Project, "Democratising Co-Design of Hardware and Control for Robot Dexterity" (03.2025 - present)
 - a. Developing high-performance tactile simulator for deep Reinforcement Learning with IsaacGym.
 - b. Investigating and developing a dexterous bimanual in-hand manipulation system with tactile sensing with IsaacGym.
2. Euro-Horizon Research Project, "Advancing the physical intelligence and performance of roBOTs towards human-like bi-manual objects MANipulation" (01.2023 - 03.2025)
 - a. Proposing a theoretical framework for a dual-arm tactile robot system called "Bi-Touch," including modeling and derivation of deep reinforcement learning reward functions.
 - b. Developing a simulation for the dual-arm tactile robot, and a hardware platform for real-world application.

- c. Learning various dual-arm robot manipulation tasks in a simulation environment through deep reinforcement learning and apply them to real-world environments through the technique of virtual-to-real transfer, including resetting and organizing object poses, and handling fragile objects with flexibility.
 - d. Developing a ROS2-based control framework for pose-and-shear tactile servoing using a 7-DoF robotic arm.
3. Leverhulme Trust Research Project, “A biomimetic forebrain for robot touch” (12.2021 - 01.2023)
- a. Proposed a new concept: robotic tactile saliency, based on the human attention of touch mechanism
 - b. Developed a highly automated approach for collecting tactile saliency training data
 - c. Successfully applied the proposed concept and approach to i) position tactile servo control based on deep learning, ii) image tactile servo control based on deep reinforcement learning
 - d. Improved the robot's operational robustness, safety, and efficiency through the proposed approach
4. Engineering and Physical Sciences Research Council “Deep Reinforcement Learning for Robot” (12.2020 - 12.2021)
- a. Studied tactile robot behaviour learning based on deep reinforcement learning
 - b. Built a tactile robot simulation environment suitable for mainstream tactile sensors such as TacTip and GelSight
 - c. Applied learned robot control strategies from the simulation environment to real-world scenarios
 - d. Improved the learning efficiency and manipulation safety of tactile robots
5. Guangdong Provincial Major Science and Technology Project, “Key Technologies Research and Application of Human-Machine Collaboration and Multi-Robot Collaboration Based on Deep Reinforcement Learning” (06.2019 - 07.2020)
- a. studied robot intelligent manipulation and autonomous learning, employing deep deterministic policy gradient for optimizing collaboration strategies, and further exploring state-action space to achieve more robust strategies
 - b. Proposed two novel data augmentation methods (i) Kaleidoscope Experience Replay exploits reflectional symmetries and (ii) Goal-augmented Experience Replay which takes advantage of lax goal definitions, enhancing the adaptability and effectiveness of robot learning in a wide range of robotic manipulation tasks
6. Guangdong Provincial Major Science and Technology Project, “Development and Industrialization of Collaborative Robots for the 3C Industry Production Line” (07. 2018 - 06.2019)
- a. Researched autonomous learning systems for robot anomaly repair in human-robot collaboration, and trained robots using dynamic motion primitive algorithms to acquire normal skills and anomaly recovery skills
 - b. Utilized higher-level representation transfer and Q-learning method to enhance anomaly recovery strategy learning capabilities, enabling autonomous learning within the anomaly recovery system
7. National Natural Science Foundation of China-Guangdong Province Joint Fund Key Project “Key Theories and Methods for High-speed Coordination of Multiple Robots in Electronic Manufacturing” (07. 2016 - 10.2018)
- a. Researched high-speed coordinated collision-free motion planning and control for multiple robots, proposing a task-oriented planning method for adaptively optimizing motion paths
 - b. Contributed to developing an intelligent operation system platform for dual SCARA robots

Awards (Selected)

1. University of Bristol-China Scholarship Council joint-funded scholarship (CSC), full funding for 4 years, 2020 - 2024.
2. Best Workshop Poster Nomination at the 5th UK Robot Manipulation Workshop 2024
3. Best Workshop Poster Award at the NeurIPS 1st Touch Processing Workshop 2023
4. Best Student Paper Nomination at the 27th IEEE International Conference on Automation and Computing
5. 2018 Top-notch Innovative Personnel in Guangdong University of Technology (top 4.7% out of 359).
6. 2018 The First Prize Scholarship of Master in Guangdong University of Technology (top 7.2% out of 359)
7. 2017 Top-notch Innovative Personnel in Guangdong University of Technology (top 5.7% out of 359)
8. 2017 The First Prize Scholarship of Master in Guangdong University of Technology (top 6.7% out of 359)
9. 2016 Excellent Postgraduate Candidates Exempt from Admission Exam (top 2.9% out of 810).

Activities (Selected)

1. A paper is accepted after peer-review in the 6th UK Robot Manipulation Workshop at the King's College London (KCL), UK, from January 7 to 8, 2025, introducing the research work on NeuralTouch.
2. Visited the LASR lab at Technische Universität Dresden in Germany from June 10 to 12, 2024, where I presented all research work conducted during my doctoral period and engage in academic discussions with Professor Roberto Calandra and his students.
3. Attended the 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2024) in Yokohama, Japan, from May 11 to 18, 2024, where I am invited to give a presentation on the paper "Bi-Touch: Bimanual Tactile Manipulation with Sim-to-Real Deep Reinforcement Learning".
4. Participated in the 5th UK Robot Manipulation Workshop at the University of Oxford, UK, from January 11 to 12, 2024, introducing and demonstrating the research work on "Tactile Saliency Prediction".
5. Attended the ICRA 2024 Workshop on "Robot Embodiment through Visuo-Tactile Perception" in London, UK, on June 2 2023, presenting my work on "Attention of Robot Touch for Robust Sim-to-Real Tactile Control".
6. Presented a research report on "Sim-to-real Deep Reinforcement Learning for Bimanual Tactile Manipulation" at The 4th UK Robot Manipulation Workshop at the University of Bristol, UK, from January 12 to 13, 2023.
7. Presented my accepted paper "Tactile Gym 2.0: Sim-to-Real Deep Reinforcement Learning for Comparing Low-Cost High-Resolution Robot Touch" at The 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2022) in Kyoto, Japan, October 23 to 27, 2022.
8. Showcased research on dexterous robot manipulation based on deep learning and tactile sensing at the 5th Conference on Robot Learning (CoRL) in London, UK, from November 8 to 11, 2021.

Teaching and Advising

1. In 2025, I supervised a summer intern Abdulahi Jama at the University of Bristol on developing bimanual in-hand manipulation with irregular objects.
2. Since 2024, I have been co-supervising a PhD student, Bowen Deng, together with Nathan F. Lepora at the University of Bristol on deep learning-based tactile sensing for tool use.
3. Between 2021 and 2024, I co-supervised a total of 7 MSc students with Nathan F. Lepora at the University of Bristol, guiding them to successfully complete their final projects and dissertations. Their names and thesis titles are as follows:
 - a. Boang Guan: Combination of Tactile Gym and Mixed Reality Technique
 - b. Di Wu: Tactile Manipulation Using Deep Reinforcement Learning on an Educational Robotics Platform
 - c. Shipeng Xiong: Data-Efficient Imitation Learning for Dexterous Manipulation
 - d. Pengyuan Wei: Dual-arm Insertion with Tactile Sensing via Deep Reinforcement Learning
 - e. Jiangfeng Fan: Towards more Learning Efficiency with Educational Robot in Tactile Gym
 - f. Bowen Deng: Coarse-to-fine Robotic Pushing with Touch, Vision and Proprioception
 - g. Bosco Zhao: Collision-Free Robotic Pushing Using Artificial Potential Field and Deep Reinforcement Learning with Tactile Sensing
4. Teaching Assistant (2023) for the "Robotics Research Technology and Methods" (master's unit) at the University of Bristol, responsibilities included:
 - a. Assisting the unit director in administrative duties and providing technical support during seminars and

- laboratory sessions.
- b. Conducting weekly critical thinking and academic discussions on robotics literature.
 - c. Offering guidance, assistance, assessment, and grading for student coursework and projects.

Professional Service

1. I serve as an early-career co-organizer for the upcoming ViTac Workshop at ICRA 2026. ([ViTac at ICRA 2025](#))
2. I serve as a Program Committee in the 2025 IEEE International Conference on Robotics and Biomimetics.
3. I have been serving as a reviewer for the following journals and conferences.
 - IEEE Transactions on Robotics (T-RO)
 - IEEE/ASME Transactions on Mechatronics (T-Mech)
 - IEEE Robotics and Automation Letters (RAL)
 - Robotics and Autonomous Systems (RAS)
 - Conference on Robot Learning (CoRL)
 - Robotics: Science and Systems (RSS)
 - IEEE International Conference on Intelligent Robots and Systems (IROS)
 - IEEE International Conference on Robotics and Automation (ICRA)